

Toward Satellite-Derived Runoff Models: Evidence from Urban Heat Islands

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Abstract

Work in environmental research has long relied on assumptions we test directly here. The measurements reported here were collected over twelve independent trials under controlled conditions. Neither of the alternative formulations improved on the simple additive model in our setting. Residual error grows approximately linearly with the number of coupled parameters.

1. Introduction

We release the processing scripts and intermediate artefacts to support independent verification. Taken together, these findings argue for treating the two regimes separately in future analyses. Longer observation windows narrow the confidence interval but do not shift the central estimate. Residual error grows approximately linearly with the number of coupled parameters.

The reduced model retains most of the explanatory power at a fraction of the computational cost. Longer observation windows narrow the confidence interval but do not shift the central estimate. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Taken together, these findings argue for treating the two regimes separately in future analyses. We release the processing scripts and intermediate artefacts to support independent verification. Residual error grows approximately linearly with the number of coupled parameters. Replicate agreement was high, suggesting the protocol is robust to small handling differences. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Taken together, these findings argue for treating the two regimes separately in future analyses. The measurements reported here were collected over twelve independent trials under controlled conditions. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

Longer observation windows narrow the confidence interval but do not shift the central estimate. Calibration drift over the trial period remained below the threshold set out in the protocol. Residual error grows approximately linearly with the number of coupled parameters. The reduced model retains most of the explanatory power at a fraction of the computational cost. Neither of the alternative formulations improved on the simple additive model in our setting.

2. Related Work

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3. Materials and Methods

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4. Experimental Setup

Residual error grows approximately linearly with the number of coupled parameters. The reduced model retains most of the explanatory power at a fraction of the computational cost. Taken together, these findings argue for treating the two regimes separately in future analyses. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. We release the processing scripts and intermediate artefacts to support independent verification.

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5. Results

Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The reduced model retains most of the explanatory power at a fraction of the computational cost.

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6. Discussion

We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Calibration drift over the trial period remained below the threshold set out in the protocol. We release the processing scripts and intermediate artefacts to support independent verification. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Neither of the alternative formulations improved on the simple additive model in our setting. Replicate agreement was high, suggesting the protocol is robust to small handling differences.

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